

Fructose and galactose metabolism

1. A 10-year-old girl is brought to the physician for a follow up examination. She had no issues in the past. Urinalysis done the previous week showed presence of a reducing sugar in the urine that was negative for glucose oxidase. A repeat urinalysis shows a reducing sugar that is not glucose. An enzyme deficiency disorder is confirmed by further laboratory studies. Which of the following enzymatic reactions is most likely impaired in this patient?

- A. Phosphorylation of galactose
- B. Cleavage of fructose 1-phosphate
- C. Formation of UDP-galactose from galactose 1-phosphate
- D. Phosphorylation of fructose
- E. Formation of glucose from glucose 6-phosphate

2. A 20-day-old girl is brought to the physician by her parents because of a 4-day history of poor feeding and failure to gain weight. Physical examination shows jaundice and hepatomegaly. There is also milestone developmental delay. Urinalysis shows a reducing sugar that is not glucose. Which of the following intermediates is most likely accumulating in her liver?

- A. Fructose
- B. Fructose 1-phosphate
- C. Glucose 6-phosphate
- D. Galactose 1-phosphate
- E. Fructose 1,6-bisphosphate
- F. Sorbitol

3. A 4-year-old boy is brought to the physician by his mother because of a 1-month history of tripping over objects. Ophthalmoscopic examination shows bilateral lens opacification. Urinalysis shows a reducing sugar that is negative for glucose oxidase test. Laboratory studies liver enzymes (ALT and AST) and serum bilirubin within normal limits. Which of the following enzymes is most likely deficient in this patient?

- A. Galactose 1-phosphate uridyl transferase (GALT)
- B. Fructokinase
- C. Aldolase B

D. Galactokinase

E. Lactase

4. A 9-month-old boy is brought to the emergency department by his parents because of a 2-hour history of drowsiness. The mother says that there were similar episodes in the past week. The mother says that she has given him fruit puree about 3 hours ago. Urinalysis shows the presence of reducing sugar that is negative for glucose oxidase test. Serum ALT and AST are elevated. Which of the following disorders is most likely in the infant?

A. von Gierke disease

B. Classical galactosemia

C. Essential fructosuria

D. Hereditary fructose intolerance

E. Intestinal disaccharidase deficiency

F. Non classical galactosemia

5. A 3-year-old girl is brought to the physician by her mother because of a 2-week history of tripping over objects. Ophthalmoscopic examination shows bilateral lens opacification. Urinalysis shows a reducing sugar that is negative for glucose oxidase test. Liver enzymes and serum bilirubin are within normal limits. Which of the following enzymes is most likely responsible for the formation of cataracts in the lens?

A. Galactose 1-phosphate uridyl transferase

B. Aldose reductase

C. Sorbitol dehydrogenase

D. Hexokinase

E. Aldolase B

F. Galactokinase

Compare galactosemia, lactose intolerance, essential fructosuria, hereditary fructose intolerance

	Enzyme deficient and accumulated substrate	Predominant clinical features	Biochemical basis for clinical features	Biochemical basis for management
Classical Galactosemia				
Non-classical Galactosemia				
Lactose intolerance				
Essential fructosuria				
Hereditary fructose intolerance				

Answer Key:

1. D
2. D
3. D
4. D
5. B

Short explanations:

1. A 10-year-old girl is brought to the physician for a follow up examination. She had no issues in the past. Urinalysis done the previous week showed presence of a reducing sugar in the urine that was negative for glucose oxidase. A repeat urinalysis shows a reducing sugar that is not glucose. An enzyme deficiency disorder is confirmed by further laboratory studies. Which of the following enzymatic reactions is most likely impaired in this patient?

Comment: This is a child with essential fructosuria. Note the benign and asymptomatic nature of the disorder – most often detected by routine urine analysis. No treatment necessary; can avoid dietary fructose and sucrose.

A. Phosphorylation of galactose – This is catalyzed by galactokinase; Deficiency of galactokinase causes cataracts

B. Cleavage of fructose 1-phosphate – This reaction is catalyzed by aldolase B; deficiency of this enzyme results in hereditary fructose intolerance characterized by hypoglycemia and liver damage

C. Formation of UDP-galactose from galactose 1-phosphate – This reaction is catalyzed GALT (galactose 1-phosphate uridyl transferase; Deficiency of this enzyme results in classical galactosemia characterized by liver damage, cataracts and intellectual developmental delay

D. Phosphorylation of fructose – **CORRECT answer - This reaction is catalyzed by fructokinase; Deficiency results in benign or essential fructosuria**

E. Formation of glucose from glucose 6-phosphate – This reaction is catalyzed by glucose 6-phosphatase; deficiency results in von Gierke disease (Glycogen storage disease type I). Children have severe hypoglycemia and hepatomegaly

2. A 20-day-old girl is brought to the physician by her parents because of a 4-day history of poor feeding and failure to gain weight. Physical examination shows jaundice and hepatomegaly. There is also milestone developmental delay. Urinalysis shows a reducing sugar that is not glucose. Which of the following intermediates is most likely accumulating in her liver?

Comment: This is a child with classical galactosemia with liver involvement (jaundice and hepatomegaly) and developmental delay due to accumulation of **galactose 1-phosphate** and galactitol (cataracts and liver damage); Dietary lactose and galactose avoidance is advised; Stop breast feeding immediately and substitute galactose and lactose-free formula.

A. Fructose – found in urine in benign fructosuria and hereditary fructose intolerance

B. Fructose 1-phosphate – accumulates in hepatocytes in hereditary fructose intolerance

C. Glucose 6-phosphate – accumulates in von Gierke disease

D. Galactose 1-phosphate – **CORRECT ANSWER – accumulates in classical galactosemia**

E. Fructose 1,6-bisphosphate – intermediate of glycolysis and gluconeogenesis

F. Sorbitol – accumulates in lens of patients with hyperglycemia (uncontrolled diabetics)

3. A 4-year-old boy is brought to the physician by his mother because of a 1-month history of tripping over objects. Ophthalmoscopic examination shows bilateral lens opacification. Urinalysis shows a reducing sugar that is negative for glucose oxidase test. Laboratory studies liver enzymes (ALT and AST) and serum bilirubin within normal limits. Which of the following enzymes is most likely deficient in this patient?

Comment: This is a child with non-classical galactosemia with only cataracts; Liver function tests and intellectual development are normal; Dietary galactose and lactose restriction is advised.

A. Galactose 1-phosphate uridyl transferase (GALT) – causes classical galactosemia; triad of cataracts, liver damage and developmental milestone delay

B. Fructokinase – Benign disorder with fructose in urine

C. Aldolase B – Hereditary fructose intolerance – causes hypoglycemia and liver damage

D. Galactokinase – **CORRECT ANSWER - deficiency of galactokinase; children have cataracts due to galactitol formation in lens**

E. Lactase – children have severe diarrhea; bloating and abdominal cramps; no reducing sugar in urine; no liver damage or developmental delay or cataracts

4. A 9-month-old boy is brought to the emergency department by his parents because of a 2-hour history of drowsiness. The mother says that there were similar episodes in the past week. The mother says that she has given him fruit puree about 3 hours ago. Urinalysis shows the presence of reducing sugar that is negative for glucose oxidase test. Serum ALT and AST are elevated. Which of the following disorders is most likely in the infant?

Comment: This is a child with hereditary fructose intolerance; Note the hypoglycemic episodes following (trigger) fructose (fruit) in the diet. Also, with prolonged exposure to fructose, liver function is deranged (elevated ALT and AST levels); Dietary avoidance of fructose and sucrose is advised.

A. von Gierke disease – children have severe fasting hypoglycemia, and hepatomegaly. Liver enzymes are usually elevated. Glycogen accumulation in liver. Avoid fasting and administer corn starch to prevent hypoglycemia.

B. Classical galactosemia - Deficiency of GALT results in classical galactosemia characterized by liver damage, cataracts and intellectual developmental delay; Some patients may also have hypoglycemia due to phosphate trapping and ATP deficiency. Galactose and lactose restricted diet is advised.

C. Essential fructosuria - Benign disorder with fructose in urine; No major organ damage

D. Hereditary fructose intolerance - **CORRECT ANSWER - causes hypoglycemia and liver damage a few hours after sucrose or fructose ingestion. Managed by avoiding fructose and sucrose in the diet.**

E. Intestinal disaccharidase deficiency – also known as Lactose intolerance - children have severe diarrhea; bloating and abdominal cramps; no reducing sugar in urine; no liver damage or developmental delay or cataracts; Normal serum liver enzymes. Lactose-free diet is advised.

F. Non classical galactosemia - Deficiency of galactokinase causes cataracts; Dietary lactose and galactose restriction is advised

5. A 3-year-old girl is brought to the physician by her mother because of a 2-week history of tripping over objects. Ophthalmoscopic examination shows bilateral lens opacification. Urinalysis shows a reducing sugar that is negative for glucose oxidase test. Liver enzymes and serum bilirubin are within normal limits. Which of the following enzymes is most likely responsible for the formation of cataracts in the lens?

Comment: This is a child with non-classical galactosemia with lens opacity (cataracts) – the enzyme responsible for the cataracts is aldose reductase in the lens. **Aldose reductase** converts galactose to galactitol (sugar alcohol) in the lens. Galactitol is osmotically active and holds water in the lens resulting in lens opacity and cataracts

A. Galactose 1-phosphate uridyl transferase (GALT) – Patients with GALT deficiency also have cataracts due to galactitol formation in the lens. The question is testing the enzyme which is forming galactitol from galactose.

B. Aldose reductase – **CORRECT ANSWER – action of aldose reductase forms the polyol (alcohol) galactitol which increases the water content of lens resulting in cataracts. The mechanism of cataract formation is similar in GALT deficiency and galactokinase deficiency.**

C. Sorbitol dehydrogenase – not found in lens. In the liver and seminal vesicle, it converts sorbitol to fructose

D. Hexokinase – converts glucose to glucose 6-phosphate. First reaction of glycolysis and is involved in glucose trapping within cells

E. Aldolase B – deficiency of this enzyme results in hereditary fructose intolerance characterized by severe hypoglycemia following sucrose or fructose in the diet.

F. Galactokinase – the patient has galactokinase deficiency and as a result they have high blood galactose levels. Galactose is converted to galactitol by aldose reductase. Galactitol is responsible for the cataracts in children with untreated galactosemia (classical and non-classical)

Compare galactosemia, lactose intolerance, essential fructosuria, hereditary fructose intolerance

	Enzyme deficient and accumulated substrate	Predominant clinical features	Biochemical basis for clinical features	Biochemical basis for management
Classical Galactosemia	GALT; Galactose 1-phosphate accumulates; Galactitol in lens, liver and brain	Jaundice (Liver damage), cataracts and developmental delay (triad)	Accumulation of galactose 1-P for liver and brain; Galactitol accumulation - cataracts	Avoid lactose and galactose in diet
Non-classical Galactosemia	Galactokinase	Cataracts	Galactitol accumulation - cataracts	Avoid lactose and galactose in diet
Lactose intolerance	Lactase	Diarrhea, bloating (no reducing sugar in urine)	Intestinal undigested lactose is degraded by bacterial action into gases	Lactose avoidance in diet
Essential fructosuria	Fructokinase	Benign – detected by reducing sugar in urine	Benign	Fructose may be avoided in the diet
Hereditary fructose intolerance	Aldolase B; Fructose 1-phosphate accumulates	Hypoglycemia, jaundice	Trapping of Pi – results in ATP deficiency and inhibition of glycogenolysis and gluconeogenesis. Accumulation of fructose 1-P and depletion of Pi – liver disease	Fructose and sucrose avoidance in diet

Practice Questions: Fatty Acid Synthesis

1. A group of researchers are observing metabolic changes in a group of women aged 30 years. They are all provided a meal containing 55-60% calories from carbohydrates. It is observed that the liver is actively synthesizing fatty acids after the meal. Which of the following is the activated end product of this process in the liver of these women?

- A. Pyruvate
- B. Citrate
- C. Acetyl CoA
- D. Palmitate
- E. Palmitoyl CoA

2. A 26-year old woman comes to the physician for a routine examination. She had delivered a healthy boy two-months ago. She exclusively breastfeeding her child. After a carbohydrate rich meal, apart from the liver, which other organ or tissue would be a major site of fatty acid synthesis in the woman?

- A. Kidneys
- B. Adipose tissue
- C. Mammary gland
- D. Heart
- E. Lungs

3. A researcher is observing metabolic changes in a group of 60-year-old men after a meal. They are all given a carbohydrate rich meal (containing 55-60% carbohydrates). After the meal, it is observed that the liver is actively synthesizing fatty acids. Which of the following enzymes most likely provides reducing equivalent for this biosynthetic pathway?

- A. Pyruvate dehydrogenase complex
- B. Glyceraldehyde 3-phosphate dehydrogenase
- C. Malic enzyme (cytosolic malate dehydrogenase)
- D. Glycerol kinase
- E. Glycerol-3-phosphate dehydrogenase

4. A group of researchers are studying absorption of dietary fats in a group of healthy adolescents aged 15 years. They are all given a balanced meal containing 25% of calories from fats and 60% calories from carbohydrates. It is observed that triacylglycerol is actively synthesized in the intestinal mucosal cells after the meal. Which of the following pathways is most likely active in these cells for the synthesis of these macromolecules?

- A. Monoacylglycerol (MAG) pathway
- B. Carnitine Shuttle
- C. Glycolysis
- D. Glycerol-3-phosphate pathway
- E. Pentose phosphate pathway

5. A researcher is observing metabolism in a group of healthy 30-year old men. It is observed that fatty acid synthesis is active in the liver after a carbohydrate rich meal. It is observed that C-20 fatty acids are synthesized from palmitic acid in the endoplasmic reticulum. Which of the following enzymes would most likely be involved in this process in the endoplasmic reticulum?

- A. Desaturase
- B. Elongase
- C. Acetyl CoA carboxylase
- D. Fatty acid synthase
- E. Malic enzyme

Answer Key

1 = E

2 = C

3 = C

4 = A

5 = B

Short explanations:

1. A group of researchers are observing metabolic changes in a group of women aged 30 years. They are all provided a meal containing 55-60% calories from carbohydrates. It is observed that the liver is actively synthesizing fatty acids after the meal. Which of the following is the activated end product of this process in the liver of these women?

- A. Pyruvate – Is the end product of glycolysis. It enters mitochondria to be converted to acetyl CoA by pyruvate dehydrogenase complex
- B. Citrate – Is a regulator of acetyl CoA carboxylase. It activates acetyl CoA carboxylase and facilitates the polymerization of the enzyme. Acetyl CoA formed in the mitochondria is transported to the cytosol for fatty acid and cholesterol synthesis as citrate.
- C. Acetyl CoA – Is the starting substrate for fatty acid synthesis and cholesterol synthesis. It forms Malonyl CoA by acetyl CoA carboxylase.
- D. Palmitate – Is the end product of fatty acid synthesis. It is a 16-C fatty acid
- E. **Palmitoyl CoA – CORRECT answer** – Palmitate is the end product of fatty acid synthesis. It is then activated to palmitoyl CoA for TAG formation. The activation to palmitoyl CoA requires Thiokinase (same enzyme used for the activation of fatty acids for beta-oxidation). Palmitoyl CoA allosterically inhibits acetyl CoA carboxylase.

2. A 26-year old woman comes to the physician for a routine examination. She had delivered a healthy boy two-months ago. She exclusively breastfeeding her child. After a carbohydrate rich meal, apart from the liver, which other organ or tissue would be a major site of fatty acid synthesis in the woman?

- A. Kidneys
- B. Adipose tissue – Fatty acids are generally synthesized in the liver and sent to the adipose tissue in VLDL. In the endothelium of the adipocytes, lipoprotein lipase cleaves the TAG to form fatty acids which are taken up by adipocytes to be later stored as TAG in adipocytes.
- C. **Mammary gland – CORRECT ANSWER** – Fatty acid synthesis is active in the lactating mammary gland until the infant is breast fed.
- D. Heart
- E. Lungs

3. A researcher is observing metabolic changes in a group of 60-year-old men after a meal. They are all given a carbohydrate rich meal (containing 55-60% carbohydrates). After the meal, it is observed that the liver is actively synthesizing fatty acids. Which of the following enzymes most likely provides reducing equivalent for this biosynthetic pathway?

- A. Pyruvate dehydrogenase complex – Forms acetyl CoA from pyruvate. It is a mitochondrial enzyme; Requires NAD⁺, FAD, CoA, TPP and lipoic acid
- B. Glyceraldehyde 3-phosphate dehydrogenase – Enzyme of glycolysis; Forms NADH; Inhibited by arsenic; Forms the high energy compound 1,3-Bisphosphoglycerate
- C. **Malic enzyme (cytosolic malate dehydrogenase) – CORRECT ANSWER** – Malic enzyme in the cytosol which converts malate to pyruvate is also a source of reducing equivalents (NADPH) for fatty acid synthesis. NADPH is also provided by the glucose 6-phosphate DH from the pentose phosphate pathway.
- D. Glycerol kinase – Converts glycerol to glycerol 3-P. It is present in liver and absent in adipocytes. Hence, adipocytes cannot form TAG from glycerol. Adipocytes use DHAP from glycolysis for TAG formation
- E. Glycerol-3-phosphate dehydrogenase – Is an enzyme of the glycerol P shuttle which transports NADH from the cytosol into the mitochondrial matrix as FADH₂. In adipocytes and liver, the enzyme converts dihydroxyacetone phosphate (DHAP) into glycerol 3-phosphate for TAG formation.

4. A group of researchers are studying absorption of dietary fats in a group of healthy adolescents aged 15 years. They are all given a balanced meal containing 25% of calories from fats and 60% calories from carbohydrates. It is observed that triacylglycerol is actively synthesized in the intestinal mucosal cells after the meal. Which of the following pathways is most likely active in these cells for the synthesis of these macromolecules?

- A. **Monoacylglycerol (MAG) pathway – CORRECT ANSWER** – Dietary TAG is hydrolyzed by pancreatic lipase to form monoacylglycerol (MAG). MAG is absorbed into the mucosal cells with the help of bile salt micelles. Within the intestinal mucosal cell, TAG is resynthesized from MAG. The TAG formed are later assembled with proteins and phospholipids and released as chylomicrons into the lymphatics of the intestine
- B. Carnitine Shuttle – Carnitine is used for transport of fatty acids into the mitochondrial matrix for beta-oxidation of fatty acids

- C. Glycolysis – Glycolysis forms DHAP for the formation of glycerol 3-P in the liver and adipocytes for TAG synthesis
- D. Glycerol-3-phosphate pathway – Glycolysis forms DHAP for the formation of glycerol 3-P in the liver and adipocytes for TAG synthesis
- E. Pentose phosphate pathway – Forms NADPH (reducing equivalents) for fatty acid synthesis and cholesterol synthesis

5. A researcher is observing metabolism in a group of healthy 30-year old men. It is observed that fatty acid synthesis is active in the liver after a carbohydrate rich meal. It is observed that C-20 fatty acids are synthesized from palmitic acid in the endoplasmic reticulum. Which of the following enzymes would most likely be involved in this process in the endoplasmic reticulum?

- A. Desaturase – Is used for the formation of double bonds in fatty acids; Remember, that Linoleic acid and alpha-linolenic acid are dietary essential fatty acids and cannot be synthesized in humans.
- B. **Elongase – CORRECT ANSWER** – Adds 2 C-units to the newly synthesized palmitic acid (16 C) to form the other longer chain fatty acids in the endoplasmic reticulum
- C. Acetyl CoA carboxylase – Regulated enzyme of fatty acid synthesis; Allosterically activated by citrate which also favors polymerization; And inhibited by palmitoyl CoA (end product) which favors monomer formation. Glucagon and epinephrine inhibit acetyl CoA carboxylase by phosphorylating the enzyme via AMP-Kinase. Insulin activates acetyl CoA carboxylase by dephosphorylating the enzyme via activating protein phosphatase.
- D. Fatty acid synthase – Multienzyme complex required for fatty acid synthesis. Palmitic acid (16 C) is the end product formed.
- E. Malic enzyme – Cytosolic enzyme which provides NADPH for fatty acid synthesis

Practice questions: Adipose tissue lipolysis, Beta oxidation and ketone body metabolism

1. A 20-year-old man is brought to the physician because of a 1-day history of weakness. He has been fasting for 2 days to lose weight. Laboratory studies show elevated circulating ketone body levels. Which of the following enzymes would most likely be active in the liver in this patient?
 - A. Mitochondrial HMG CoA synthase
 - B. HMG CoA reductase
 - C. Acetyl CoA carboxylase
 - D. Succinyl CoA: Acetoacetate CoA transferase
 - E. Hormone sensitive lipase
2. A 2-month-old girl is brought to the physician because of poor feeding and failure to gain weight since birth. Physical examination shows jaundice, hepatomegaly and a delay in the

developmental milestones. Laboratory studies show Zellweger syndrome. Which of the following processes is most likely impaired in this patient?

- A. Degradation of sphingomyelin
- B. Oxidation of very long chain fatty acids
- C. Ketone body synthesis
- D. Beta-oxidation of long chain fatty acids
- E. Arachidonic acid formation from linoleic acid

3. A 9-month-old boy is brought to the emergency department with a 1-hour history of seizures. He has had a 3-day history of respiratory infection and is feeding poorly. Laboratory studies show an inherited deficiency of medium chain acyl CoA dehydrogenase (MCAD). Which of the following laboratory findings is most likely in this patient?

- A. Elevated circulating phytanic acid levels
- B. Ketonuria and ketoacidosis
- C. Elevated circulating C-22 and C-24 fatty acids
- D. Low level of circulating ketone bodies
- E. Hyperglycemia

4. A researcher is observing metabolic adaptations in a group of 30-year-old women. They are all advised to fast for 3 days and metabolic changes are observed. On the 3rd day of the fast, it is observed that the circulating ketone bodies are elevated in all patients. Which of the following best describes these intermediates?

- A. Methylmalonyl-CoA is an intermediate of ketogenesis
- B. Synthesized predominantly in the kidney
- C. Skeletal muscle utilizes glucose in preference to ketone bodies as an energy source during fasting
- D. Produced by the oxidation of glucogenic amino acids in the fasting state
- E. Are weak acids and reduce serum bicarbonate levels

5. A 25-year-old woman is brought to the emergency department by her husband in a comatose state. She has a 15-year history of type 1 diabetes. She has been irregular with her insulin treatment in the past week. Urinalysis shows the presence of ketone bodies and glucose. Which of the following metabolic changes is most likely observed in this patient on admission?

- A. Arterial pH is elevated
- B. 3-hydroxybutyrate levels are low
- C. Hormone sensitive lipase is phosphorylated
- D. Beta-oxidation in the liver is inhibited
- E. Liver gluconeogenesis is inhibited

6. A 10-month-old boy is brought to the emergency department with a 6-hour history of lethargy and drowsiness. He has a 2-day history of fever, vomiting and diarrhea and has been refusing feeds. Laboratory studies show hypoglycemia and ketone bodies are undetectable. Additional laboratory studies show elevated C-8 and C-10 acyl carnitine in blood and urine. Which of the following is the most likely diagnosis?

- A. Myopathic beta-oxidation disorder

- B. Systemic beta-oxidation disorder
- C. Muscle glycogen storage disorder
- D. Liver glycogen storage disorder
- E. Classical galactosemia
- F. Alpha-oxidation defect

7. A 1-year-old girl is brought to the emergency department by her parents because of a 1-hour history of seizures. She has been identified to have an inherited deficiency of medium chain acyl CoA dehydrogenase (MCAD) by newborn screening. Which of the following is most likely observed in the patient on admission?

- A. Ketone body production is decreased
- B. Blood glucose level is increased
- C. β -oxidation in liver is increased
- D. Liver glycogenolysis is defective
- E. Lipolysis in adipose tissue is low

8. A 20-year-old man comes to the physician because of a 1-month history of muscle cramps while doing sustained exercise. Laboratory studies show normal plasma glucose levels. Liver biopsy studies are normal. A muscle biopsy shows normal glycogen stores but increased lipid droplets. Which of the following enzymes is most likely deficient in this patient?

- A. Carnitine palmitoyl transferase-I (CPT-I)
- B. Carnitine palmitoyl transferase-II (CPT-II)
- C. Muscle glycogen phosphorylase
- D. Liver glycogen phosphorylase
- E. Medium chain acyl CoA dehydrogenase (MCAD)

9. A researcher studying metabolism observes that liver cytosolic malonyl CoA levels were elevated during a particular metabolic state. Which of the following pathways are most likely active in the liver during this state?

- A. Glycogenolysis
- B. Beta-oxidation of long chain fatty acids
- C. Gluconeogenesis
- D. Formation of acetoacetate and 3-hydroxybutyrate
- E. Triacylglycerol formation and fatty acid synthesis

10. A 1-year-old girl is brought to the physician by her parents for a follow up examination. She had three episodes of hypoglycemia and low ketone body levels following episodes of illness. Which of the following supplements is most likely useful in this patient?

- A. Carnitine
- B. Citrate

- C. N-acetyl glutamate
- D. Benzoic acid
- E. Creatine

Answer Key:

- 1. A
- 2. B
- 3. D
- 4. E
- 5. C
- 6. B
- 7. A
- 8. B
- 9. E
- 10. A

Short explanations:

1. A 20-year-old man is brought to the physician because of a 1-day history of weakness. He has been fasting for 2 days to lose weight. Laboratory studies show elevated circulating ketone body levels. Which of the following enzymes would most likely be active in the liver in this patient?
 - A. **Mitochondrial HMG CoA synthase – CORRECT ANSWER** – Regulatory enzyme of ketogenesis. It is found only in hepatocytes and is active during ketogenesis in the fasting state. Note that HMG CoA lyase is also only found in the liver
 - B. HMG CoA reductase – Is less active in the fasting state. This is the rate limiting enzyme of cholesterol biosynthesis
 - C. Acetyl CoA carboxylase - Is less active in the fasting state. This is the rate limiting enzyme of fatty acid biosynthesis
 - D. Succinyl CoA: Acetoacetate CoA transferase – This enzyme is NOT present in the liver. It is present only in extrahepatic tissues. It is active in the fasting state for utilization of the ketone bodies in muscle. Also known as Thiophorase
 - E. Hormone sensitive lipase – Is active in the fasting state but is present in the adipose tissue and NOT in the liver
2. A 2-month-old girl is brought to the physician because of poor feeding and failure to gain weight since birth. Physical examination shows jaundice, hepatomegaly and a delay in the developmental milestones. Laboratory studies show Zellweger syndrome. Which of the following processes is most likely impaired in this patient?

- A. Degradation of sphingomyelin – Is defective in children with Neiman-Pick disease due to deficiency of sphingomyelinase. It is a lysosomal storage disorder
 - B. **Oxidation of very long chain fatty acids – CORRECT ANSWER** – fatty acids of chain length more than 22-C atoms are oxidized in the peroxisomes. In Zellweger syndrome, the peroxisomes are not formed (Peroxisomal biogenesis disorder) and all peroxisomal functions are affected
 - C. Ketone body synthesis – takes place in the mitochondria
 - D. Beta-oxidation of long chain fatty acids – takes place in the mitochondria
 - E. Arachidonic acid formation from linoleic acid – The elongation and desaturation take place in the endoplasmic reticulum
3. A 9-month-old boy is brought to the emergency department with a 1-hour history of seizures. He has had a 3-day history of respiratory infection and is feeding poorly. Laboratory studies show an inherited deficiency of medium chain acyl CoA dehydrogenase (MCAD). Which of the following laboratory findings is most likely in this patient?
- A. Elevated circulating phytanic acid levels – Observed in patients with Refsum disease (defect in alpha-oxidation)
 - B. Ketonuria and ketoacidosis – Children would have impaired beta-oxidation resulting in low levels of acetyl CoA which is a substrate for beta-oxidation; Ketone body levels would be low
 - C. Elevated circulating C-22 and C-24 fatty acids – is seen in patients with Zellweger syndrome. Peroxisomal defect characterized by a defect in oxidation of very-long chain fatty acids
 - D. **Low level of circulating ketone bodies – CORRECT ANSWER** – Due to less acetyl CoA formed as a result of reduced rates of beta oxidation
 - E. Hyperglycemia – Children with MCAD deficiency would have hypoglycemia due to reduced rates of gluconeogenesis – there is less of ATP formed for gluconeogenesis and acetyl CoA levels are low – resulting in reduced activation of pyruvate carboxylase; Also, peripheral tissues are utilizing glucose as an energy source rather than fatty acids
4. A researcher is observing metabolic adaptations in a group of 30-year-old women. They are all advised to fast for 3 days and metabolic changes are observed. On the 3rd day of the fast, it is observed that the circulating ketone bodies are elevated in all patients. Which of the following best describes these intermediates?
- A. Methylmalonyl-CoA is an intermediate of ketogenesis - Methylmalonyl CoA is formed as an intermediate in the conversion of propionyl CoA to succinyl CoA. HMG CoA is an intermediate of Ketogenesis
 - B. Synthesized predominantly in the kidney – They are synthesized mainly by the liver; but are utilized by the muscle (skeletal and cardiac) and by the brain (only in prolonged fasting)
 - C. Skeletal muscle utilizes glucose in preference to ketone bodies as an energy source during fasting – During fasting the low insulin glucagon ratio reduces the number of GLUT-4 in muscle, as a result there is reduced uptake of glucose by the muscle. Ketone bodies are preferentially used during the fasting state by muscle

- D. Produced by the oxidation of glucogenic amino acids in the fasting state – Majority of the ketone bodies are formed by beta oxidation of fatty acids. Remember, that oxidation of ketogenic amino acids can also form ketone bodies to a minor extent during fasting
- E. **Are weak acids and reduce serum bicarbonate levels – CORRECT answer** – 3-hydroxybutyrate and acetoacetate are weak acids. They release protons that are buffered by bicarbonate (ECF buffer) and decrease serum bicarbonate levels; High levels of ketone bodies can result in metabolic acidosis with increased anion gap. Ketone bodies are unmeasured anions and contribute to anion gap.

5. A 25-year-old woman is brought to the emergency department by her husband in a comatose state. She has a 15-year history of type 1 diabetes. She has been irregular with her insulin treatment in the past week. Urinalysis shows the presence of ketone bodies and glucose. Which of the following metabolic changes is most likely observed in this patient on admission?
Comment: Type 1 diabetics have a risk of hyperglycemia and ketosis when they miss their insulin dose, or when insulin supplement is not sufficient or during periods of illness.

- A. Arterial pH is elevated – Arterial pH will be low (acidosis) due to excessive ketone body levels in the blood; results in high anion gap metabolic acidosis
- B. 3-hydroxybutyrate levels are low – 3-hydroxybutyrate levels are elevated and is responsible for the metabolic acidosis
- C. **Hormone sensitive lipase is phosphorylated – CORRECT ANSWER** – Due to insulin deficiency, hormone sensitive lipase is in the phosphorylated state and has increased activity resulting in uncontrolled lipolysis. As a result, fatty acid levels in circulation are elevated
- D. Beta-oxidation in the liver is inhibited – Increased free fatty acid levels in circulation will increase the rates of beta oxidation
- E. Liver gluconeogenesis is inhibited – In uncontrolled diabetes mellitus, liver gluconeogenesis is very active resulting in elevated blood glucose levels and hyperglycemia

6. A 10-month-old boy is brought to the emergency department with a 6-hour history of lethargy and drowsiness. He has a 2-day history of fever, vomiting and diarrhea and has been refusing feeds. Laboratory studies show hypoglycemia and ketone bodies are undetectable. Additional laboratory studies show elevated C-8 and C-10 acyl carnitine in blood and urine. Which of the following is the most likely diagnosis?

- A. Myopathic beta-oxidation disorder – Patients present with muscle cramps during exercise and aerobic exercise intolerance
- B. **Systemic beta-oxidation disorder – CORRECT ANSWER** – Beta oxidation in the liver is defective resulting in reduced rates of acetyl CoA formation (acetyl CoA is the substrate for ketone body synthesis); Hypoglycemia is due to peripheral tissues using more amounts of glucose due to a defect in beta oxidation. Also, beta oxidation provides ATP (energy) and acetyl CoA to activate pyruvate carboxylase of gluconeogenesis – a defect in gluconeogenesis results in less of acetyl CoA available to activate gluconeogenesis; and gluconeogenesis is not occurring at optimal rates resulting in severe hypoglycemia

- C. Muscle glycogen storage disorder – Patients present with muscle cramps during exercise and anaerobic/ ischemic exercise intolerance
- D. Liver glycogen storage disorder – Present with hypoglycemia but ketone body production is not affected; Also have lactic acidosis, hyperuricemia.
- E. Classical galactosemia – Present with hypoglycemia few hours after ingestion of milk; liver involvement (jaundice); cataracts and developmental delay
- F. Alpha-oxidation defect – Patients with Refsum disease have a defect in peroxisomal alpha-oxidation of branched chain fatty acids. Present with visual and neurological defects

7. A 1-year-old girl is brought to the emergency department by her parents because of a 1-hour history of seizures. She has been identified to have an inherited deficiency of medium chain acyl CoA dehydrogenase (MCAD) by newborn screening. Which of the following is most likely observed in the patient on admission?

- A. **Ketone body production is decreased - CORRECT ANSWER** – Beta oxidation in the liver is defective resulting in reduced rates of acetyl CoA formation (acetyl CoA is the substrate for ketone body synthesis); Hypoglycemia is due to peripheral tissues using more amounts of glucose due to a defect in beta oxidation. Also, beta oxidation provides ATP (energy) and acetyl CoA to activate pyruvate carboxylase of gluconeogenesis – a defect in beta-oxidation results in less of acetyl CoA available to activate gluconeogenesis; and gluconeogenesis is not occurring at optimal rates resulting in severe hypoglycemia
- B. Blood glucose level is increased – MCAD deficiency results in hypoglycemia
- C. β -oxidation in liver is increased – beta oxidation is reduced due to MCAD deficiency. MCAD is an enzyme of the beta oxidation pathway
- D. Liver glycogenolysis is defective – Liver gluconeogenesis is affected due to less of ATP formed via beta oxidation and less of acetyl CoA for stimulation of pyruvate carboxylase
- E. Lipolysis in adipose tissue is low – Hormone sensitive lipase is active in these children, but there is a defect in beta oxidation

8. A 20-year-old man comes to the physician because of a 1-month history of muscle cramps while doing sustained exercise. Laboratory studies show normal plasma glucose levels. Liver biopsy studies are normal. A muscle biopsy shows normal glycogen stores but increased lipid droplets. Which of the following enzymes is most likely deficient in this patient?

Comment: This is a patient with defect in generation of energy for muscle contraction during aerobic exercise. This patient most likely has a myopathic beta oxidation defect. Additional laboratory findings are elevated serum creatine kinase-MM (CK-MM) levels. Blood lactate levels increase following ischemic exercise, indicating glycogen breakdown to glucose 6-phosphate and glycolysis to form lactate is not affected. These patients are able to perform anaerobic exercise!

- A. Carnitine palmitoyl transferase-I (CPT-I) – The liver isoform is most commonly affected resulting in systemic beta oxidation defect. Presents with hypoglycemia and hypoketosis

- B. **Carnitine palmitoyl transferase-II (CPT-II) – Correct answer** - The muscle isoform of CPT-II is most commonly deficient resulting in myopathic beta oxidation disorder. Muscle TAG (Lipid droplets) are increased
- C. Muscle glycogen phosphorylase – In McArdle disease, patients have muscle cramps but there is increased glycogen accumulation in the muscle. Patients with McArdle cannot tolerate anaerobic activity
- D. Liver glycogen phosphorylase – In Hers disease, patients have mild hypoglycemia and glycogen accumulation in the liver
- E. Medium chain acyl CoA dehydrogenase (MCAD) – Most often presents as the systemic form with fasting hypoglycemia and hypoketosis

9. A researcher studying metabolism observes that liver cytosolic malonyl CoA levels were elevated during a particular metabolic state. Which of the following pathways are most likely active in the liver during this state?

Comment: Cytosolic malonyl CoA levels are high when fatty acid biosynthesis is active (Fed state; high insulin glucagon ratio). Elevated cytosolic malonyl CoA inhibits CPT-I and reduces the transports of fatty acids into mitochondria for beta-oxidation.

- A. Glycogenolysis – active during fasting and low insulin : glucagon ratio. High glucagon levels during fasting, results in increased cAMP, active protein kinase A, glycogen phosphorylase is in the phosphorylated form and active.
- B. Beta-oxidation of long chain fatty acids – active during fasting and low insulin: glucagon ratio. Malonyl CoA inhibits CPT-I and inhibits transport of fatty acids into the mitochondria for beta oxidation
- C. Gluconeogenesis – active during fasting and low insulin: glucagon ratio
- D. Formation of acetoacetate and 3-hydroxybutyrate – active during fasting and low insulin: glucagon ratio
- E. **Triacylglycerol formation and fatty acid synthesis – CORRECT ANSWER** – Liver glycolysis is active and forms dihydroxyacetone phosphate (DHAP) which is used for TAG synthesis in the liver. Acetyl CoA carboxylase is active when the insulin: glucagon ratio is high. Acetyl CoA carboxylase (rate limiting enzyme of fatty acid synthesis) forms malonyl CoA.

10. A 1-year-old girl is brought to the physician by her parents for a follow up examination. She had three episodes of hypoglycemia and low ketone body levels following episodes of illness. Which of the following supplements is most likely useful in this patient?

Comment: This is a child with systemic beta oxidation disorder (hypoglycemia and hypoketosis). This question can be answered by eliminating the other choices and identifying the best response.

- A. **Carnitine – CORRECT ANSWER** – Carnitine supplementation has been tried and improvement of symptoms is seen in systemic beta oxidation and myopathic beta oxidation disorders

- B. Citrate – Is an activator of fatty acid biosynthesis (acetyl CoA carboxylase)
- C. N-acetyl glutamate – Is an allosteric activator of CPS-I (carbamyl phosphate synthetase-I) of the urea cycle. It is used in patients with urea cycle disorders to increase activity of CPS-I.
- D. Benzoic acid – Benzoic acid combines with glycine to form hippuric acid. Administration of benzoic acid reduces the blood ammonia levels in patients with urea cycle disorders
- E. Creatine – Creatine as creatine phosphate is a storage form of energy in muscle. It provides ATP for muscle contraction during the first few seconds of muscle contraction

Summary of fatty acid oxidation disorders:

	Alpha oxidation	Very long chain fatty acid oxidation	Myopathic fatty acid oxidation defect	Systemic beta oxidation defect
Disorder	Refsum disease	Zellweger syndrome	CPT-II deficiency	MCAD deficiency/ primary carnitine deficiency
Enzyme defect/ organelle involved	Alpha oxidation/ Phytanate alpha hydroxylase/ peroxisome	Peroxisomal very long chain fatty acid oxidation and all peroxisomal functions	Affects muscle beta oxidation; CPT-II deficiency impairs fatty acid transport into mitochondria	MCAD deficiency, primary carnitine deficiency, or CPT-I deficiency results in defective beta oxidation in liver / mitochondria
Accumulating substrate	Phytanic acid (branched chain fatty acids)	Very long chain fatty acids (C22-C24)	Increased intramuscular TAG (Lipid droplets in muscle)	C8-C10 acyl carnitine
Clinical features	Blindness	Presents at birth – intellectual disability, hepatomegaly	Muscle cramps on exercise; exercise intolerance for prolonged exercise	Hypoglycemia 6-8 hours after the last meal; episodes on stress/ infection
Lab findings	Increased branched chain fatty acids	Increased Very long chain fatty acids (C22-C24) – peroxisomes not well formed	Muscle biopsy – shows increased TAG and low carnitine levels in carnitine deficiency	C8-C10 acyl carnitine; Hypoglycemia; low ketone body levels; dicarboxylic acids

Practice Questions: Regulation of Fatty Acid Synthesis and Cholesterol Metabolism

1. A group of researchers are observing metabolic changes in a group of 20-year-old women. They are all given a carbohydrate rich meal (50% calories from carbohydrates) and liver metabolism is observed one hour later. It is observed that the hepatocytes are actively synthesizing cholesterol and there is increased activity of the rate limiting enzyme of cholesterol synthesis. Which of the following enzymes is most likely being observed by the researcher?

- A. Citrate lyase
- B. Acetyl CoA carboxylase
- C. HMG CoA synthase
- D. HMG CoA reductase
- E. 7-dehydrocholesterol reductase

2. A research group is studying cholesterol synthesis in the liver in experimental rats. Hepatic cytosolic cholesterol levels are reduced in the animals and their effect on the rate limiting enzyme of the pathway is observed. Which of the following effects is most likely observed on the rate limiting enzyme of this pathway under these conditions?

- A. Inhibition of the enzyme
- B. Down regulation of the enzyme
- C. Early degradation of the enzyme
- D. Increased transcription of the enzyme
- E. Phosphorylation of the enzyme

3. Metabolic changes are observed in the liver in a group of 20-year-old men. They are all given a meal containing 60% calories from carbohydrate (carbohydrate rich meal) and their liver metabolism is observed. It is observed that fatty acid and cholesterol synthesis are active in the liver one hour after the meal. Which of the following enzymes is most likely required for both the pathways?

- A. Fatty acid synthase
- B. Acetyl CoA carboxylase
- C. Citrate lyase
- D. HMG CoA Synthase
- E. HMG CoA Reductase

4. Metabolic changes are observed in a group of 40-year-old men after giving them a meal containing 60% carbohydrates (carbohydrate rich meal). It is observed that fatty acid synthesis is active after the meal. Which of the following intermediates most likely allosterically activates the rate limiting enzyme of this pathway?

- A. Citrate
- B. Cholesterol
- C. Palmitoyl CoA
- D. AMP-Kinase
- E. AMP

5. A research group is studying metabolic regulation in experimental animals. It is observed that during the synthesis of fatty acids, one of the intermediates of the pathway ensures that fatty acid oxidation does not take place at the same time by inhibiting an enzyme involved in transport of long chain fatty acids for β -oxidation. Which of the following intermediates most likely performs this function?

- A. Palmitoyl CoA
- B. Mevalonate
- C. Citrate
- D. Acetyl CoA
- E. Malonyl CoA

Regulation of Fatty Acid Synthesis and Cholesterol Metabolism: Answer key and explanations.

Answer Key

- 1 = D
- 2 = D
- 3 = C
- 4 = A
- 5 = E

Short Explanations:

1. A group of researchers are observing metabolic changes in a group of 20-year-old women. They are all given a carbohydrate rich meal (50% calories from carbohydrates) and liver metabolism is observed one hour later. It is observed that the hepatocytes are actively synthesizing cholesterol and there is increased activity of the rate limiting enzyme of cholesterol synthesis. Which of the following enzymes is most likely being observed by the researcher?

- A. Citrate lyase - This is a cytosolic enzyme which converts [citrate and CoA](#) to oxaloacetate and [acetyl-CoA](#), linking [carbohydrate metabolism](#) to [fatty acid biosynthesis](#), where acetyl-CoA is used as a precursor.
- B. Acetyl CoA carboxylase - Acetyl CoA carboxylase (ACC) is the rate-limiting enzyme in fatty acid biosynthesis, but not for cholesterol synthesis.
- C. HMG CoA synthase - catalyzes the reaction in which [acetyl-CoA](#) condenses with [acetoacetyl-CoA](#) to form [3-hydroxy-3-methylglutaryl-CoA](#) (HMG-CoA). HMG-CoA is an intermediate in both [cholesterol](#)

[synthesis](#) and [ketogenesis](#). There are 2 forms of the enzyme: [Mitochondrial HMG CoA synthase is used for ketogenesis](#) and [Cytosolic HMG CoA synthase is used for cholesterol synthesis](#)

- D. **HMG CoA reductase – CORRECT ANSWER.** HMG-CoA reductase is the rate limiting enzyme for cholesterol synthesis. It is inhibited by statins.
- E. 7-dehydrocholesterol reductase - 7-Dehydrocholesterol reductase is responsible for the final step in the production of cholesterol. Deficiency of 7-Dehydrocholesterol reductase results in Smith-Lemli-Opitz Syndrome (SLOS).

2. A research group is studying cholesterol synthesis in the liver in experimental rats. Hepatic cytosolic cholesterol levels are reduced in the animals and their effect on the rate limiting enzyme of the pathway is observed. Which of the following effects is most likely observed on the rate limiting enzyme of this pathway under these conditions?

- A. Inhibition of the enzyme - Incorrect. Low levels of cytosolic cholesterol does not inhibit HMG CoA reductase.
- B. Down regulation of the enzyme - Incorrect. Low levels of cytosolic cholesterol does not downregulate HMG CoA reductase.
- C. Early degradation of the enzyme - Incorrect
- D. **Increased transcription of the enzyme – CORRECT ANSWER.** Low cytosolic cholesterol levels stimulate the release of a regulatory (SREBP) protein from the endoplasmic reticulum (ER), resulting in increased transcription of HMG CoA reductase gene and subsequent increase in the amount of HMG CoA reductase in the liver.
- E. Phosphorylation of the enzyme - Incorrect. Low levels of cytosolic cholesterol does not lead to phosphorylation of HMG CoA reductase. Phosphorylation of the enzyme results in lower activity of the enzyme

3. Metabolic changes are observed in the liver in a group of 20-year-old men. They are all given a meal containing 60% calories from carbohydrate (carbohydrate rich meal) and their liver metabolism is observed. It is observed that fatty acid and cholesterol synthesis are active in the liver one hour after the meal. Which of the following enzymes is most likely required for both the pathways?

- A. Fatty acid synthase - Although active during the well fed state, fatty acid synthase is involved in fatty acid synthesis, but not in cholesterol synthesis. Multifunctional enzyme which uses NADPH as reducing equivalents for fatty acid synthesis.
- B. Acetyl CoA carboxylase - This enzyme is involved in fatty acid synthesis but not in cholesterol synthesis. It is the rate limiting enzyme of fatty acid synthesis. Contains biotin as a coenzyme
- C. Citrate lyase – **CORRECT ANSWER.** After a carbohydrate rich meal, glycolysis in liver forms pyruvate which is next converted to acetyl CoA by pyruvate dehydrogenase complex in the mitochondria. Mitochondrial acetyl CoA concentration is high. Acetyl CoA condenses with oxaloacetate and forms citrate by citrate synthase. Citrate is transported from the mitochondria into the cytosol. In the cytosol, citrate lyase cleaves [citrate](#) to oxaloacetate and [acetyl-CoA](#), linking [carbohydrate metabolism](#) to [fatty acid biosynthesis](#) and cholesterol metabolism. Acetyl-CoA is used as a precursor for both fatty acid and cholesterol synthesis.
- D. HMG CoA Synthase - is required to cholesterol synthesis and ketogenesis, but not to fatty acid synthesis. Cholesterol synthesis uses the cytosolic isoenzyme; Whereas, ketogenesis uses the mitochondrial enzyme. Mitochondrial HMG CoA synthase is the rate limiting enzyme of ketogenesis in the liver.

- E. HMG CoA Reductase - is the rate limiting enzyme and regulated enzyme of cholesterol synthesis. It converts HMG-CoA to mevalonate and uses NADPH. It is inhibited by statins. HMG CoA reductase is NOT used for fatty acid synthesis.

4. Metabolic changes are observed in a group of 40-year-old men after giving them a meal containing 60% carbohydrates (carbohydrate rich meal). It is observed that fatty acid synthesis is active after the meal. Which of the following intermediates most likely allosterically activates the rate limiting enzyme of this pathway?

- A. **Citrate – CORRECT ANSWER.** Citrate shifts the equilibrium towards formation of polymers of the rate limiting enzyme (acetyl CoA carboxylase), thereby activating the enzyme.
- B. Cholesterol – Elevated cholesterol inhibits de-novo cholesterol synthesis by inhibiting transcription of HMG-CoA reductase.
- C. Palmitoyl CoA - The end product of fatty acid synthesis is palmitate which is then activated to palmitoyl CoA. Palmitoyl CoA facilitates the formation of monomers of acetyl CoA carboxylase that are less active. It also acts as an allosteric inhibitor of the enzyme.
- D. AMP-Kinase: AMP-activated protein kinase (AMPK) is activated by AMP, glucagon and epinephrine. It phosphorylates and inhibits acetyl CoA carboxylase.
- E. AMP – AMP is adenosine monophosphate. AMP activates AMP-Kinase which results in phosphorylation and conversion of acetyl CoA carboxylase to the inactive form.

5. A research group is studying metabolic regulation in experimental animals. It is observed that during the synthesis of fatty acids, one of the intermediates of the pathway ensures that fatty acid oxidation does not take place at the same time by inhibiting an enzyme involved in transport of long chain fatty acids for β -oxidation. Which of the following intermediates most likely performs this function?

- A. Palmitoyl CoA - The end product of fatty acid synthesis is palmitate which is then activated to palmitoyl CoA. Palmitoyl CoA and is then used for TAG synthesis. Palmitoyl CoA allosterically inhibits acetyl CoA carboxylase (fatty acid synthesis)
- B. Mevalonate - Mevalonate is an intermediate formed during cholesterol synthesis. HMG-CoA is converted to mevalonate by HMG-CoA reductase (regulated enzyme of cholesterol synthesis).
- C. Citrate - Citrate serves as a means of transporting acetyl CoA from the mitochondria into the cytosol. In the cytosol, citrate is cleaved to release oxaloacetate and cytosolic acetyl CoA. Citrate allosterically activates acetyl CoA carboxylase (fatty acid synthesis)
- D. Acetyl CoA - Cytosolic acetyl CoA is a precursor (starting product) required to synthesize fatty acids.
- E. **Malonyl CoA – CORRECT ANSWER:** Malonyl-CoA, which is formed in the cytosol during fatty acid synthesis, inhibits CPT-I which is involved in the transport of long-chain fatty acids into the mitochondria for β -oxidation. This ensures that during fatty acid synthesis, β -oxidation is inhibited. Newly synthesized FA remains in the cytosol and are not immediately oxidized in the fed state.

Cholesterol Metabolism

Practice Questions

1. Metabolic changes are observed in a group of healthy 20-year-old women. They are all given a meal containing 60% calories from carbohydrates (carbohydrate rich) and liver metabolism is observed one hour

later. It is observed that the hepatocytes are actively synthesizing cholesterol. Which of the following products is most likely formed by the enzyme catalyzing the rate limiting reaction of this pathway?

- A. Citrate
- B. Acetyl CoA
- C. HMG CoA
- D. Acetoacetyl CoA
- E. Squalene
- F. Mevalonate
- G. Malonyl CoA
- H. Farnesyl pyrophosphate
- I. Isopentenyl pyrophosphate
- J. Geranyl pyrophosphate

2. A 2-week-old girl is brought to the physician by her parents for a follow-up examination. She has microcephaly and has had poor weight gain since birth. Laboratory studies confirm Smith-Lemli-Opitz Syndrome. Formation of which of the following compounds is most likely impaired in this patient?

- A. Palmitic acid
- B. Cholesterol
- C. Acetoacetate and 3-hydroxybutyrate
- D. Cholic acid and chenodeoxycholic acid
- E. Sphingomyelin

3. Cholesterol synthesis in the liver is observed in a group of healthy 30-year-old men. It is observed that hepatocytes are actively synthesizing cholesterol after a carbohydrate rich meal. Which of the following coenzymes is most likely required for this process?

- A. Pyridoxal phosphate (PLP)
- B. NADP+
- C. NAD
- D. NADPH
- E. FADH₂
- F. Biotin

4. After a meal rich in fatty acids and carbohydrates, conjugated primary bile salts are being actively synthesized in the liver in a healthy 15-year old boy. Which of the following molecules most likely belongs to this group?

- A. Cholic acid
- B. Glycocholic acid
- C. Chenodeoxycholic acid
- D. Lithocholic acid
- E. Deoxycholic acid

5. Hepatic bile acid synthesis is observed in a 40-year-old man. It is observed that bile acids are synthesized by a series of reactions in the liver. Which of the following enzymes most likely catalyzes the first step in this pathway?

- A. HMG CoA synthase
- B. HMG CoA reductase
- C. HMG CoA lyase
- D. 7 α -hydroxylase
- E. Thiolase

6. A 2-week-old boy is brought to the physician by his parents because of poor feeding and poor weight gain. Physical examination shows microcephaly. Laboratory studies confirm Smith-Lemli-Opitz syndrome. Targeted gene mutation analysis shows a loss of function mutation in a gene encoding an enzyme. Which of the following enzymes is most likely affected in this patient?

- A. HMG CoA reductase
- B. Thiophorase
- C. 7-dehydrocholesterol reductase
- D. 7 α -hydroxylase
- E. Citrate lyase

Answer Key

1 = F

2 = B

3 = D

4 = B

5 = D

6 = C

Short explanations:

Cholesterol Metabolism

Practice Questions

1. Metabolic changes are observed in a group of healthy 20-year-old women. They are all given a meal containing 60% calories from carbohydrates (carbohydrate rich) and liver metabolism is observed one hour later. It is observed that the hepatocytes are actively synthesizing cholesterol. Which of the following products is most likely formed by the enzyme catalyzing the rate limiting reaction of this pathway?

- A. Citrate - Is the transport form of acetyl CoA from the mitochondria to the cytosol. Cytosolic citrate lyase forms acetyl CoA in the cytosol which is then used for fatty acid and cholesterol synthesis
- B. Acetyl CoA - Is the starting material for both fatty acid and cholesterol synthesis in the cytosol. Acetyl CoA in the mitochondria can be used by the Krebs cycle, or for ketone body synthesis.
- C. HMG CoA - Is formed in the cytosol in cholesterol synthesis by cytosolic HMG CoA synthase. The mitochondrial HMG CoA synthase forms HMG CoA for ketone body synthesis in the liver
- D. Acetoacetyl CoA – Is an intermediate of ketogenesis and cholesterol synthesis. It next forms HMG CoA by HMG CoA synthase. Cytosolic HMG CoA is used for cholesterol synthesis and mitochondrial HMG CoA is used for ketogenesis
- E. Squalene - Is a 30C intermediate of cholesterol synthesis. Formed by the condensation of 2 molecules of Farnesyl pyrophosphate (15 C intermediate)
- F. **Mevalonate – CORRECT ANSWER – HMG CoA reductase forms Mevalonate from HMG CoA. It requires NADPH from pentose phosphate pathway. This enzyme is also inhibited by statins.**
- G. Malonyl CoA - Is an intermediate of fatty acid synthesis. It is formed from acetyl CoA by Acetyl CoA carboxylase which is the rate limiting and regulated enzyme of fatty acid synthesis
- H. Farnesyl pyrophosphate – Is a 15-C intermediate of cholesterol synthesis pathway
- I. Isopentenyl pyrophosphate – Is a 5-C intermediate of cholesterol synthesis pathway
- J. Geranyl pyrophosphate – Is a 10-C intermediate of cholesterol synthesis pathway

2. A 2-week-old girl is brought to the physician by her parents for a follow up examination. She has microcephaly and has poor weight gain since birth. Laboratory studies confirm Smith-Lemli-Opitz Syndrome. Formation of which of the following compounds is most likely impaired in this patient?

- A. Palmitic acid – Is the end product of fatty acid synthesis. It is a 16-C fatty acid. It is activated to palmitoyl CoA and later used for TAG synthesis. It can also be used for synthesis of longer chain fatty acids and unsaturated fatty acids in the endoplasmic reticulum.
- B. **Cholesterol – Correct answer - Patients with Smith-Lemli-Opitz syndrome (SLOS) have reduced cholesterol synthesis due to a deficiency of 7-dehydrocholesterol reductase. They are supplemented with dietary cholesterol. However, remember that cholesterol cannot cross the blood brain barrier and cannot improve the microcephaly. Note that the development of brain is impaired even during embryonic life in these children.**
- C. Acetoacetate and 3-hydroxybutyrate – These are ketone bodies; Ketogenesis is not impaired in SLOS

- D. Cholic acid and chenodeoxycholic acid – These are primary bile acids; Synthesis of primary bile acids is not primarily affected in SLOS
- E. Sphingomyelin – Sphingomyelin synthesis is not affected. Patients with Niemann-Pick disease have sphingomyelin accumulation

3. Cholesterol synthesis in the liver is observed in a group of healthy 30-year-old men. It is observed that the hepatocytes are actively synthesizing cholesterol after a carbohydrate rich meal. Which of the following coenzymes is most likely required for this process?

- A. Pyridoxal phosphate (PLP) – Is used in amino acid metabolism
- B. NADP⁺ - Forms NADPH by the pentose phosphate pathway and malic enzyme. NADPH is used for cholesterol and fatty acid synthesis
- C. NAD – Is used for glycolysis, TCA cycle, PDH complex in beta oxidation. NADH formed enters the ETC to form 3 ATP
- D. **NADPH – Correct answer** – NADPH provides the reducing equivalents required for cholesterol and fatty acid synthesis. NADPH is provided by the pentose phosphate pathway and the cytosolic malic enzyme. NADPH is used by HMG CoA reductase (cholesterol synthesis) and by fatty acid synthase multifunctional enzyme (fatty acid synthesis).
- E. FADH₂ – Formed in beta oxidation by medium chain acyl CoA dehydrogenase (MCAD) and also used for PDH complex and succinate dehydrogenase (TCA cycle)
- F. Biotin – is present in carboxylases; Acetyl CoA carboxylase (fatty acid synthesis), pyruvate carboxylase (gluconeogenesis) and propionyl CoA carboxylase (conversion of propionyl CoA to methylmalonyl CoA)

4. After a meal rich in fatty acids and carbohydrates, conjugated primary bile salts are being actively synthesized in the liver in a healthy 15-year old boy. Which of the following molecules most likely belongs to this group?

- A. Cholic acid – Is a primary bile acid synthesized from cholesterol in the liver. Addition of glycine or taurine forms the conjugated bile acids.
- B. **Glycocholic acid – Correct answer** – Cholic acid and chenodeoxycholic acid are primary bile acids which are synthesized from cholesterol in the liver. They are conjugated with glycine and taurine to form the conjugated bile acids (AKA conjugated bile salts) glycocholic acid, taurocholic acid and glycochenodeoxycholic acid and taurochenodeoxycholic acid. Conjugation occurs in the liver and the conjugated bile acids/ conjugated bile salts are released and stored in the gall bladder. The conjugated bile salts (acids) are most useful in absorption of dietary lipids and fat soluble vitamins.
- C. Chenodeoxycholic acid – Is a primary bile acid synthesized from cholesterol in the liver. Addition of glycine or taurine forms the conjugated bile acids
- D. Lithocholic acid – Is a secondary bile acid. It is formed from chenodeoxycholic acid (primary bile acid) in the intestine by bacterial action. Secondary bile acids are generally lost in feces.
- E. Deoxycholic acid – Is a secondary bile acid. It is formed from cholic acid (primary bile acid) in the intestine by bacterial action. Secondary bile acids are generally lost in feces.

5. Hepatic bile acid synthesis is observed in a 40-year old man. It is observed that bile acids are synthesized by a series of reactions in the liver. Which of the following enzymes most likely catalyzes the first step in this pathway?

- A. HMG CoA synthase – Mitochondrial HMG CoA synthase is the regulated enzyme of ketogenesis in the liver. Cytosolic HMG CoA synthase is an enzyme of cholesterol synthesis.
- B. HMG CoA reductase – Regulated enzyme of cholesterol synthesis. Uses NADPH. Inhibited by statins
- C. HMG CoA lyase – Enzyme involved in ketogenesis in liver mitochondria
- D. **7 α -hydroxylase – CORRECT answer: 7-alpha hydroxylase is the rate limiting and regulated step in the bile acid synthesis pathway. Recollect that primary bile acids are synthesized in the liver. These undergo conjugation with glycine or taurine to form the conjugated bile acids.**
- E. Thiolase – Is the last enzyme of mitochondrial fatty acid beta oxidation. It is also used in ketogenesis for the formation of acetoacetyl CoA.

6. A 2-week-old boy is brought to the physician by his parents because of poor feeding and poor weight gain. Physical examination shows microcephaly. Laboratory studies confirm Smith-Lemli-Opitz syndrome. Targeted gene mutation analysis shows a loss of function mutation in a gene encoding an enzyme. Which of the following enzymes is most likely affected in this patient?

- A. HMG CoA reductase – Regulated enzyme of cholesterol synthesis
- B. Thiophorase – AKA succinyl CoA: acetoacetate CoA transferase; It is present in skeletal muscle, heart and brain. Not present in liver; The enzyme is involved in activation of ketone bodies and is required for ketone body utilization by these tissues.
- C. **7-dehydrocholesterol reductase – CORRECT ANSWER – Loss of function mutation results in Smith-Lemli-Opitz syndrome (SLOS). These patients have reduced cholesterol synthesis. Characterized by intellectual disability, microcephaly.**
- D. 7 α -hydroxylase – Regulated and rate limiting enzyme of bile acid synthesis
- E. Citrate lyase – Cytosolic enzyme which forms acetyl CoA required for fatty acid and cholesterol synthesis